

EE2080: Lecture 9: 21 Aug 2026

TH-DTMC $\{X_n\}_{n \geq 1}$ on a state space S with a TPM P

Chapman Kolmogorov $P\{X_{n+1}=j \mid X_1=i\} = (P^n)_{ij}$

$\pi(n)$ - unconditional pmf of X_n .

then $\pi(n+1) = \pi(n)P$.

Strong Markov Property: $P\{X_{T+k}=j \mid X_1=i_1, \dots, X_T=i\} =$
(TH-DTMC) - $= P\{X_{T+k}=j \mid X_T=i\} = (P^k)_{ij}$

T is a stopping time

Hitting & Recurrence Times

Let $\{X_n, n \in \mathbb{N}\}$ a TH-DTMC on state space S with TPM P .

For $j \in S$, let $T_j^{(0)} \triangleq 0$ and

$$T_j^{(k)} = \min\{n > T_j^{(k-1)} : X_n = j\}$$

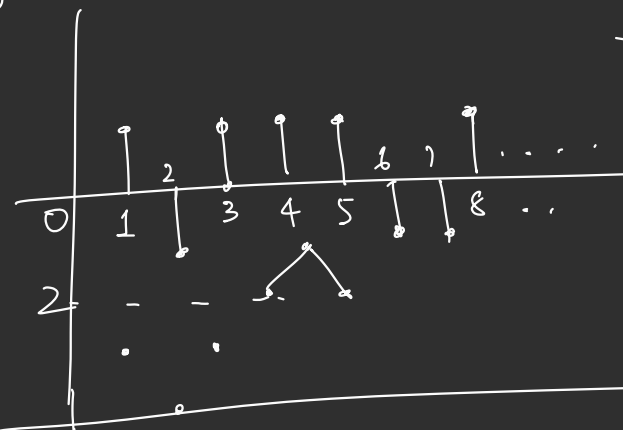
$$T_j^{(1)} = \min\{n > 0 : X_n = j\}$$

Is $T_j^{(k)}$ a stopping time?

$$T_j^{(k)} \leq n - \sum_{1 \leq i \leq n} \{X_i\} \text{ for any } k \in \mathbb{N} \text{ \& any } j \in S.$$

Strong Markov Property applies

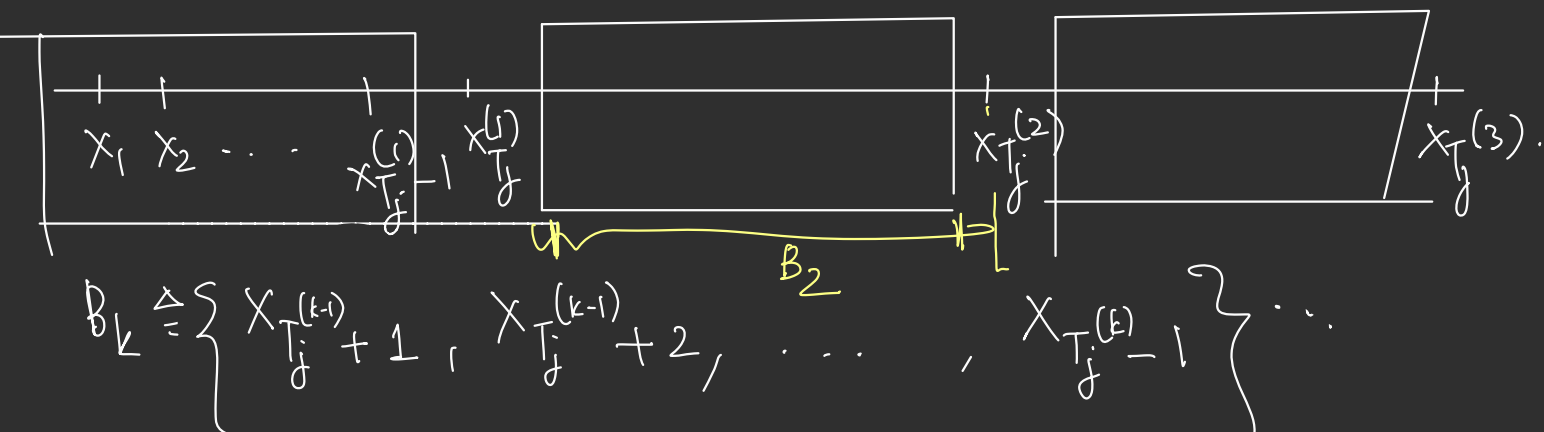
$$(X_1, \dots, X_{T_j^{(k)}-1}) \perp\!\!\!\perp (X_{T_j^{(k)}+1}, \dots) \mid X_{T_j^{(k)}}$$



$$\left. \begin{array}{l} T_2^{(1)} = 4 \\ T_2^{(2)} = 6 \\ T_2^{(3)} = 8 \end{array} \right\} 2$$

i.i.d Block Structure

: TH - DTMC $\{x_n\}_{n \geq 1}$ on S with TPM P .
 $T_j^{(k)}$ - k^{th} hitting time of state j for $k \in \mathbb{N}$, $j \in S$.



$$B_k \triangleq \{X_{T_j^{(k-1)}} + 1, X_{T_j^{(k-1)}} + 2, \dots, X_{T_j^{(k)}} - 1\} \dots$$

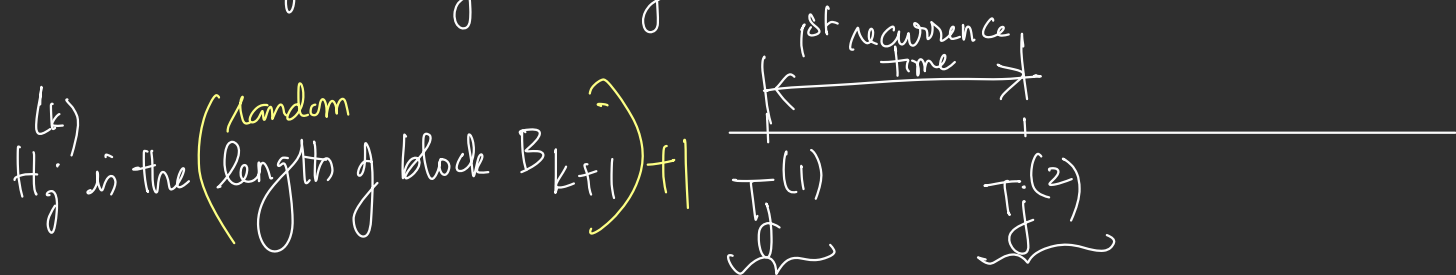
- 1) $\{B_1, B_2, B_3, \dots\}$ are independent
 2) $\{B_2, B_3, \dots\}$ are identity distributed
 of $p(T_j^{(k)} < \infty) = 1 \forall k \in \mathbb{N}$

$$(X_{T_j^{(k-1)}+1} = x_1, \dots, X_{T_j^{(k)}-1} = x)$$

For any $k \in \mathbb{N}$ $P(X_{T_j^{(k)}+1} = i) = P(X_{T_j^{(k)}} = i | X_{T_j^{(k)}} = j) = \underline{\underline{p_{ji}}}$

Recurrence Time : The k^{th} recurrence time associated with state j is

defined as $H_j^{(k)} = T_j^{(k+1)} - T_j^{(k)} \quad \forall k \in \mathbb{N}$



If $P(T_j^{(k)} < \infty) = 1 \quad \forall k \in \mathbb{N}$ then $\{H_j^{(k)}\}_{k=1}^{\infty}$ is a sequence of iid RVs.
 Follows directly from i.i.d nature of $\{B_i\}_{i \geq 2}$

First Passage Distribution.

Given that $\{X_n\}_{n \geq 0}$ is a TH-DTMC on S with TPM P .

$$f_{ij}^{(n)} = \mathbb{P}\{X_1 \neq j, X_2 \neq j, \dots, X_n = j \mid X_0 = i\} = \mathbb{P}(T_j^{(i)} = n \mid X_0 = i)$$

for any $i, j \in S$
 $n \geq 1$

$$f_{j,j}^{(n)} = \text{the prob. of first return to } j \text{ in } n \text{ steps} = \mathbb{P}(H_j^{(j)} = n)$$

$$f_{ij} = \mathbb{P}(\exists n \geq 1 \text{ s.t. } X_n = j \mid X_0 = i) = \mathbb{P}(T_j^{(i)} < \infty \mid X_0 = i)$$

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{(n)} \leq 1$$

For $i, j \in S$, $f_{ij}^{(n)}$ is called the first passage distribution for hitting j starting from state i , $f_{ij}^{(n)}$ is a PMF on the positive integers.

In general $\sum f_{ij}^{(n)} < 1 \rightarrow$ defective PMF

$f_{ij} = \sum_n f_{ij}^{(n)} < 1 \Rightarrow (1 - f_{ij}) > 0 \rightarrow$ prob. that the DTMC never visits j when it starts at i

When $f_{j,j} = 1$, we define mean recurrence time

$$\nu_{j,j} = \sum_{n=1}^{\infty} n \cdot f_{j,j}^{(n)}$$

$$f_{j,j}^{(0)} = 1 \text{ and for } i \neq j, f_{ij}^{(0)} = 0$$

(Define)

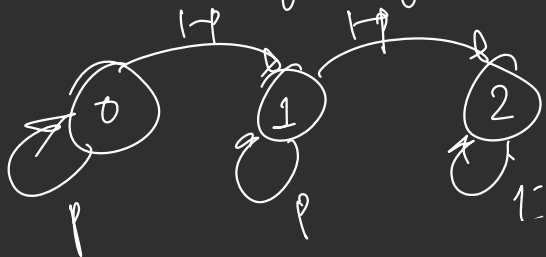
$$f_{00}^{(n)} = \mathbb{P}(T_0^{(0)} = n \mid X_0 = 0)$$

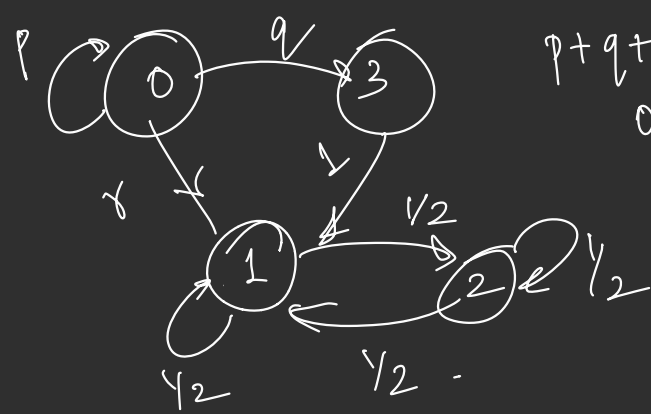
$$f_{00}^{(1)} = p$$

$$f_{00}^{(n)} = 0 \text{ for } n \geq 2$$

$$f_{ij}^{(n)} = \mathbb{P}(X_1 \neq j, \dots, X_{n-1} \neq j, X_n = j \mid X_0 = i)$$

Example:





$$p + q + r = 1$$

$$0 \leq p, q, r < 1$$

$$f_{0,1}^{(1)} = r$$

$$f_{0,1}^{(n)} =$$

$$n \geq 2$$

$$f_{0,1}^{(n)} = p^{n-1} r + p^{n-2} q$$

Find the first passage distributions of this process.

Why care about hitting times and recurrence?

- Applications:
- 1) Stock Market Analysis : How long till a stock reaches a target price?
 - 2) Analysis of Randomized Algorithms for Boolean-SAT problems with application to Electronic Design Automation (Automatic Test Pattern Gen)